HYPOTHALAMIC BULK RNA SEQUENCING

Hypothalamic response to leptin in lipodystrophy

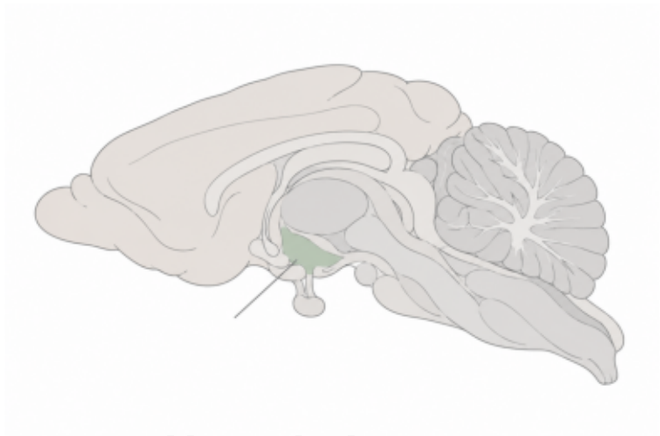
Study design and planned differential-expression analysis

A Cohorts and tissue workflow

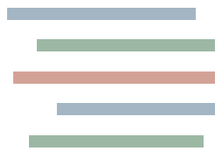
S **LD + saline**
Untreated lipodystrophy

L **LD + leptin**
Leptin-treated lipodystrophy

K **Non-LD control**
Non-lipodystrophic mice



Hypothalamus



Bulk RNA-seq
Gene-level expression

All cohorts: LDLR$^{-/-}$ background.

B Planned contrasts

L vs **S** **Leptin treatment effect**
Response within lipodystrophic mice

S vs **K** **Untreated LD phenotype**
Baseline LD-associated differences

L vs **K** **Differences after leptin**
Persistent and/or treatment-induced changes

C Joint interpretation

Compare direction and effect size

Shift toward control
LD change moves toward K

Persistent difference
LD-associated change remains

Treatment-associated change
Interpret alongside baseline LD

Analysis plan only; no expression data shown.

Positive $\log_2$FC: higher in the first group